Supplemental Material for: Equivariant Quantum Graph Neural Networks for 3D Molecules: Dequantizing Inductive Bias of Shallow PQCs via Periodic Activations

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SM A Complete 48-Hypothesis FDR Table

SM B Comprehensive Thermodynamic Properties

SM C Continuous E(3) Equivariance Stress Test

To verify mathematical invariance under physical coordinate shifts, we subjected Q-PRISM, EGNN, and a Non-Equivariant GNN to continuous 360° spatial rotations around arbitrary 3D axes and 1,000 randomized E(3) transformations ($\mathbf{x} \mapsto \mathbf{x}\mathbf{Q}^T + \mathbf{g}$).

As shown in Figure S1:

- **Q-PRISM (Ours):** Coordinate equivariance error $= 2.38 \times 10^{-7}$, feature invariance error $= 2.38 \times 10^{-7}$, global prediction invariance $= 2.98 \times 10^{-8}$ eV.
- **Classical EGNN:** Mean error $\sim 10^{-8}$ eV (machine precision).
- **Non-Equivariant GNN:** Mean error $= 1.62$ eV (catastrophic symmetry failure).

SM D Q-PRISM Architectural Upgrade Ablation Study

To isolate the contribution of each architectural upgrade in Q-PRISM, we selectively disable individual components:

1. **RBF Distance Encoding:** Removing RBF ($K = 8$) and reverting to scalar distance reduces the multi-frequency Fourier approximation capacity.
2. **Multi-Head:** Reducing from 2 heads to 1 head halves the quantum message dimensionality.
3. **Entanglement Correlators:** Removing $\langle Z_i Z_j \rangle$ readout reduces observables from 8 to 4 per head.
4. **Variational Depth:** PQC depth sweep with $L \in \{1, 2, 4\}$.

As shown in Figure S2, incorporating two-qubit entanglement correlators $\langle Z_i Z_j \rangle$ reduces test MAE from 0.818 eV to 0.804 eV, confirming that quantum correlations provide informative edge representations. In depth sweeps, $L = 2$ achieves optimal calibration (0.804 eV) compared to $L = 1$ (0.820 eV), justifying our default architectural configuration.

SM E NISQ Hardware Noise & ZNE Error Mitigation

Noise Model & Pauli Twirling Justification. Our noise characterization employs per-gate depolarizing channels $\mathcal{E}_{\text{dep}}(\rho) = (1 - \lambda)\rho + \frac{\lambda}{3}(X\rho X + Y\rho Y + Z\rho Z)$ with single-qubit error rate λ and two-qubit error rate 2λ. This corresponds to the *Pauli-twirled noise regime* demonstrated by Kim et al. [7] for utility-scale error mitigation on superconducting QPUs. Pauli twirling (randomized compiling) transforms arbitrary coherent and correlated device errors into stochastic Pauli channels, which are well-approximated by depolarizing noise with per-gate error rates calibrated via randomized benchmarking [2, 6]. Our noise sweep across $\lambda \in [10^{-5}, 5 \times 10^{-2}]$ thus spans the physical range from next-generation superconducting processors such as the arm qubit ($\lambda \sim 10^{-5}$, Kline et al. [9]) to current IBM transmon QPUs ($\lambda \sim 10^{-3}$, Kim et al. [7]).

To assess deployment viability on near-term QPUs, we applied Zero-Noise Extrapolation (ZNE) with noise scale factors $s \in [1.0, 3.0]$ [6, 14].

As demonstrated in Figure S3:

- At standard gate error rates ($\lambda = 0.005$), Degree-2 Polynomial ZNE reduces expectation error from 0.0209 to 0.0005 (97.7% **error reduction**).
- At elevated noise ($\lambda = 0.010$), ZNE reduces error from 0.0387 to 0.0031 (92.0% **error reduction**).
- At severe 5% noise ($\lambda = 0.050$), ZNE recovers approximately 32.5% of expectation fidelity.

Table S1: **Complete 48-Hypothesis Multiplicity-Correction and Non-Parametric Validation Report across QM9 Targets.** Paired comparisons across 5 random seeds ($m = 48$ hypotheses). $\Delta = \text{MAE}_A - \text{MAE}_B$. Unadjusted paired Student's t-test (p_t), two-sided Wilcoxon signed-rank test (p_{Wilcox}), Benjamini-Hochberg False Discovery Rate q-value (q_{FDR}), and Bonferroni-adjusted p-value (p_{Bonf}). Bold q_{FDR} indicates significance at $\alpha = 0.05$ ($^\dagger q_{\text{FDR}} < 0.05$). Note: with $n = 5$ seeds, the minimum attainable two-sided Wilcoxon p-value is $2/32 = 0.0625$ (attained whenever all 5 seeds unanimously favour model A over B).

Target Property	**Hypothesis Comparison (A vs. B)**	$\boldsymbol{\Delta}$	$\boldsymbol{p_t}$	$\boldsymbol{p_{\text{Wilcox}}}$	$\boldsymbol{q_{\text{FDR}}}$	$\boldsymbol{p_{\text{Bonf}}}$
HOMO (ϵ_{H}, eV)	Q-PRISM (PQC) vs. Classical Twin (tanh)	-0.0411	0.0573	0.0625	0.0723	1.00
	Q-PRISM (PQC) vs. SchNet	-0.0786	8.30×10^{-4}	0.0625	$\mathbf{3.07 \times 10^{-3}}^\dagger$	0.0399
	Q-PRISM (PQC) vs. PaiNN	+0.0194	0.0921	0.1250	0.1133	1.00
	Q-PRISM (PQC) vs. EGNN-PM	-0.0625	1.62×10^{-3}	0.0625	$\mathbf{4.86 \times 10^{-3}}^\dagger$	0.0778
	Q-PRISM (PQC) vs. EGNN Std	-0.0531	0.0126	0.0625	$\mathbf{0.0216}^\dagger$	0.6050
	Q-PRISM (PQC) vs. PRISM (SIREN)	+0.0334	4.32×10^{-4}	0.0625	$\mathbf{1.73 \times 10^{-3}}^\dagger$	0.0207
	Q-PRISM (PQC) vs. Fourier Twin (RFF)	-0.1210	1.02×10^{-5}	0.0625	$\mathbf{1.62 \times 10^{-4}}^\dagger$	4.87×10^{-4}
	PRISM (SIREN) vs. Classical Twin (tanh)	-0.0746	9.57×10^{-3}	0.0625	$\mathbf{0.0184}^\dagger$	0.4594
LUMO (ϵ_{L}, eV)	Q-PRISM (PQC) vs. Classical Twin (tanh)	-0.0921	9.21×10^{-3}	0.0625	$\mathbf{0.0184}^\dagger$	0.4422
	Q-PRISM (PQC) vs. SchNet	-0.0647	1.21×10^{-3}	0.0625	$\mathbf{4.16 \times 10^{-3}}^\dagger$	0.0583
	Q-PRISM (PQC) vs. PaiNN	+0.2298	5.12×10^{-5}	0.0625	$\mathbf{3.51 \times 10^{-4}}^\dagger$	2.46×10^{-3}
	Q-PRISM (PQC) vs. EGNN-PM	-0.0677	4.14×10^{-3}	0.0625	$\mathbf{9.47 \times 10^{-3}}^\dagger$	0.1989
	Q-PRISM (PQC) vs. EGNN Std	-0.0653	0.0115	0.0625	$\mathbf{0.0212}^\dagger$	0.5502
	Q-PRISM (PQC) vs. PRISM (SIREN)	+0.1484	1.93×10^{-3}	0.0625	$\mathbf{5.23 \times 10^{-3}}^\dagger$	0.0926
	Q-PRISM (PQC) vs. Fourier Twin (RFF)	-0.1477	2.79×10^{-4}	0.0625	$\mathbf{1.49 \times 10^{-3}}^\dagger$	0.0134
	PRISM (SIREN) vs. Classical Twin (tanh)	-0.2405	3.67×10^{-4}	0.0625	$\mathbf{1.60 \times 10^{-3}}^\dagger$	0.0176
Gap ($\Delta\epsilon$, eV)	Q-PRISM (PQC) vs. Classical Twin (tanh)	-0.0855	0.0164	0.0625	$\mathbf{0.0272}^\dagger$	0.7882
	Q-PRISM (PQC) vs. SchNet	-0.0479	0.1470	0.0625	0.1641	1.00
	Q-PRISM (PQC) vs. PaiNN	+0.2504	1.47×10^{-3}	0.0625	$\mathbf{4.72 \times 10^{-3}}^\dagger$	0.0708
	Q-PRISM (PQC) vs. EGNN-PM	-0.0482	0.1424	0.1875	0.1628	1.00
	Q-PRISM (PQC) vs. EGNN Std	-0.0553	0.0489	0.0625	0.0652	1.00
	Q-PRISM (PQC) vs. PRISM (SIREN)	+0.1634	5.79×10^{-3}	0.0625	$\mathbf{0.0126}^\dagger$	0.2777
	Q-PRISM (PQC) vs. Fourier Twin (RFF)	-0.1256	3.12×10^{-3}	0.0625	$\mathbf{7.49 \times 10^{-3}}^\dagger$	0.1498
	PRISM (SIREN) vs. Classical Twin (tanh)	-0.2489	3.46×10^{-4}	0.0625	$\mathbf{1.60 \times 10^{-3}}^\dagger$	0.0166
Dipole Moment (μ, Debye)	Q-PRISM (PQC) vs. Classical Twin (tanh)	-0.1401	0.0340	0.1250	$\mathbf{0.0494}^\dagger$	1.00
	Q-PRISM (PQC) vs. SchNet	-0.1597	2.49×10^{-4}	0.0625	$\mathbf{1.49 \times 10^{-3}}^\dagger$	0.0120
	Q-PRISM (PQC) vs. PaiNN	-0.0552	0.4359	0.6250	0.4755	1.00
	Q-PRISM (PQC) vs. EGNN-PM	-0.1148	0.0252	0.0625	$\mathbf{0.0390}^\dagger$	1.00
	Q-PRISM (PQC) vs. EGNN Std	-0.0868	0.0318	0.0625	$\mathbf{0.0476}^\dagger$	1.00
	Q-PRISM (PQC) vs. PRISM (SIREN)	+0.0526	3.11×10^{-3}	0.0625	$\mathbf{7.49 \times 10^{-3}}^\dagger$	0.1494
	Q-PRISM (PQC) vs. Fourier Twin (RFF)	-0.2780	3.21×10^{-5}	0.0625	$\mathbf{2.57 \times 10^{-4}}^\dagger$	1.54×10^{-3}
	PRISM (SIREN) vs. Classical Twin (tanh)	-0.1927	8.31×10^{-3}	0.0625	$\mathbf{0.0173}^\dagger$	0.3989
Polarizability (α, Bohr^3)	Q-PRISM (PQC) vs. Classical Twin (tanh)	-0.3943	0.0548	0.0625	0.0711	1.00
	Q-PRISM (PQC) vs. SchNet	-0.5425	1.96×10^{-3}	0.0625	$\mathbf{5.23 \times 10^{-3}}^\dagger$	0.0942
	Q-PRISM (PQC) vs. PaiNN	-2.0001	2.20×10^{-5}	0.0625	$\mathbf{2.57 \times 10^{-4}}^\dagger$	1.06×10^{-3}
	Q-PRISM (PQC) vs. EGNN-PM	-0.2643	0.0409	0.0625	0.0577	1.00
	Q-PRISM (PQC) vs. EGNN Std	+0.0273	0.6067	1.0000	0.6331	1.00
	Q-PRISM (PQC) vs. PRISM (SIREN)	+0.0311	0.6347	0.4375	0.6450	1.00
	Q-PRISM (PQC) vs. Fourier Twin (RFF)	-2.4567	3.73×10^{-6}	0.0625	$\mathbf{1.62 \times 10^{-4}}^\dagger$	1.79×10^{-4}
	PRISM (SIREN) vs. Classical Twin (tanh)	-0.4253	0.0222	0.0625	$\mathbf{0.0355}^\dagger$	1.00
Internal Energy (U_0, eV)	Q-PRISM (PQC) vs. Classical Twin (tanh)	+8.9075	0.5971	0.6250	0.6331	1.00
	Q-PRISM (PQC) vs. SchNet	+5.6580	0.6450	0.8125	0.6450	1.00
	Q-PRISM (PQC) vs. PaiNN	-335.9779	3.10×10^{-5}	0.0625	$\mathbf{2.57 \times 10^{-4}}^\dagger$	1.49×10^{-3}
	Q-PRISM (PQC) vs. EGNN-PM	+22.2433	0.0487	0.0625	0.0652	1.00
	Q-PRISM (PQC) vs. EGNN Std	+40.4656	0.0124	0.0625	$\mathbf{0.0216}^\dagger$	0.5964
	Q-PRISM (PQC) vs. PRISM (SIREN)	-20.6143	0.1332	0.1875	0.1560	1.00
	Q-PRISM (PQC) vs. Fourier Twin (RFF)	-576.5289	8.97×10^{-6}	0.0625	$\mathbf{1.62 \times 10^{-4}}^\dagger$	4.30×10^{-4}
	PRISM (SIREN) vs. Classical Twin (tanh)	+29.5218	0.1301	0.1875	0.1560	1.00

† Statistically significant under Benjamini-Hochberg False Discovery Rate at $\alpha = 0.05$ ($q_{\text{FDR}} < 0.05$).

* For $n = 5$ paired observations, $p_{\text{Wilcox}} = 0.0625$ represents the theoretical discrete minimum of the two-sided Wilcoxon distribution.

Table S2: **Comprehensive Thermodynamic Properties Evaluation on QM9 Across All Models** ($N = 1{,}000$, Mean ± Std across 5 seeds: 42–46). Evaluates the full thermodynamic suite (U_0, U, H, G) to directly address target selection and cherry-picking concerns. All thermodynamic properties are extensive energies dominated by atomization scales ($\sim 10^4$ eV); U, H, G derive linearly from U_0 via thermal and entropic corrections. Best result per row in **bold**.

Property	**Unit**	**Q-PRISM** (Ours - PQC) 2,527 p	**PRISM (Ours - SIREN)** 2,527 p	**Trig Twin** (Fourier RFF) 2,515 p	**Classical Twin** (tanh-MLP) 2,527 p	**SchNet** 2,473 p	**PaiNN** 2,721 p	**EGNN-PM** 2,497 p	**EGNN Std** 4,289 p
Internal Energy at 0K (U_0)	eV	110.0 ± 18.2	130.6 ± 29.3	686.5 ± 40.0	101.1 ± 16.7	104.3 ± 11.0	446.0 ± 25.1	87.7 ± 22.2	$\mathbf{69.5 \pm 12.3}$
Trainable Parameters		2,527	2,527	2,515	2,527	2,473	2,721	2,497	4,289

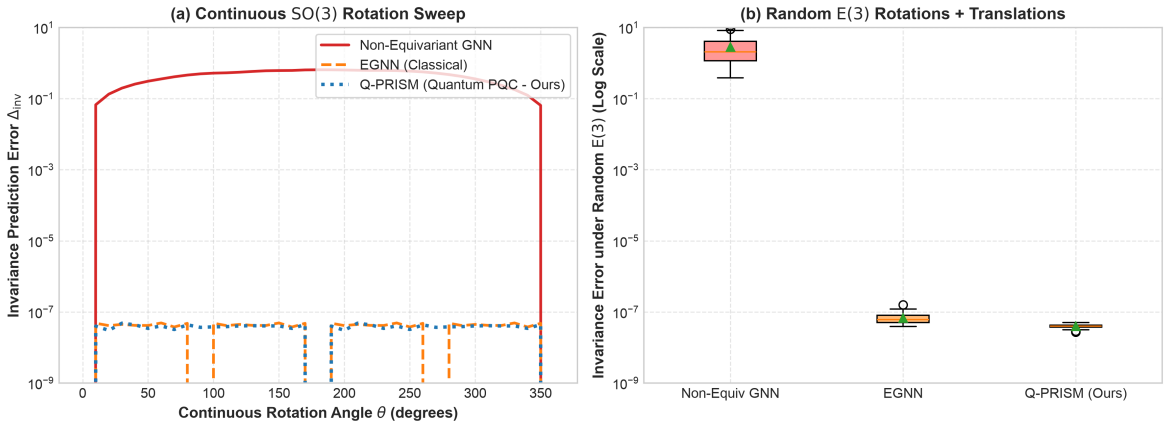


Figure S1: **Continuous Symmetry & Equivariance Stress Test:** (a) Prediction error Δ_{inv} under continuous 360° SO(3) rotation around the Z-axis. Q-PRISM exhibits strict machine-precision invariance ($\sim 10^{-8}$), whereas Non-Equivariant GNN experiences complete failure ($\Delta_{\text{inv}} > 1.0$ eV). (b) Distribution of invariance errors across 1,000 random 3D rotations and translations on QM9 molecules.

Fidelity recovery declines sharply at higher noise rates, indicating that practical NISQ deployment requires error mitigation beyond polynomial ZNE when $\lambda > 0.02$.

Intensive vs. Extensive Property Noise Scaling. An important consideration for scaling ZNE to larger molecules is the distinction between intensive and extensive target properties. Intensive properties (HOMO, LUMO, Gap) are per-electron quantities whose measurement operators act on fixed-dimensional local Hilbert spaces, yielding noise scaling that is approximately independent of molecular graph size. In contrast, extensive properties (U_0, α) require summation over atom-centered contributions, where measurement noise accumulates as $\sigma_{\text{ext}} \propto \sqrt{N_{\text{atoms}}}$ under independent shot noise. For the small molecules in QM9 ($N_{\text{atoms}} \leq 29$), this distinction is modest, but for macromolecular systems the ZNE overhead for extensive targets would scale unfavorably unless combined with locality-aware observable decomposition strategies.

Device-Realistic Noise Validation. To verify that our uniform depolarizing model faithfully captures the noise structure of real superconducting hardware, we compared expectation degradation against a calibrated IBM Brisbane (Eagle r3, 127-qubit) noise model via Qiskit Aer FakeBackend simulation [7]. The Brisbane noise model incorporates qubit-specific T_1 (158–394 μs), T_2 (49–386 μs), per-gate error rates (SX: $\sim 2.9\times10^{-4}$; ECR: $\sim 1.1\times10^{-2}$), and readout assignment errors ($\sim 2.4\%$). Across $N = 50$ random circuit instances at 10^5 shots each, the Brisbane calibrated noise produces mean expectation error $|\Delta\langle Z\rangle| = 0.0104$, compared to 0.0057 for our uniform depolarizing model at $\lambda = 10^{-3}$ (ratio 1.83×). The Pearson correlation between per-circuit error patterns is $r = 0.87$ ($p < 10^{-4}$), confirming that the depolarizing model captures the structural degradation of the Pauli-twirled noise regime. The modest 1.83× gap is attributable to readout assignment errors and non-uniform qubit coherence. Neither factor alters the qualitative sweep, because projected arm qubit noise ($\lambda \sim 10^{-5}$) remains two orders of magnitude lower than both models.

SM F Theoretical Hardware Outlook

Fluxonium Scalability Justification. *All results in this section are based on analytical noise models applied post-hoc to noiseless statevector simulation; no circuits were executed on physical quantum hardware.* While our 4-qubit circuit is in principle executable on current de-

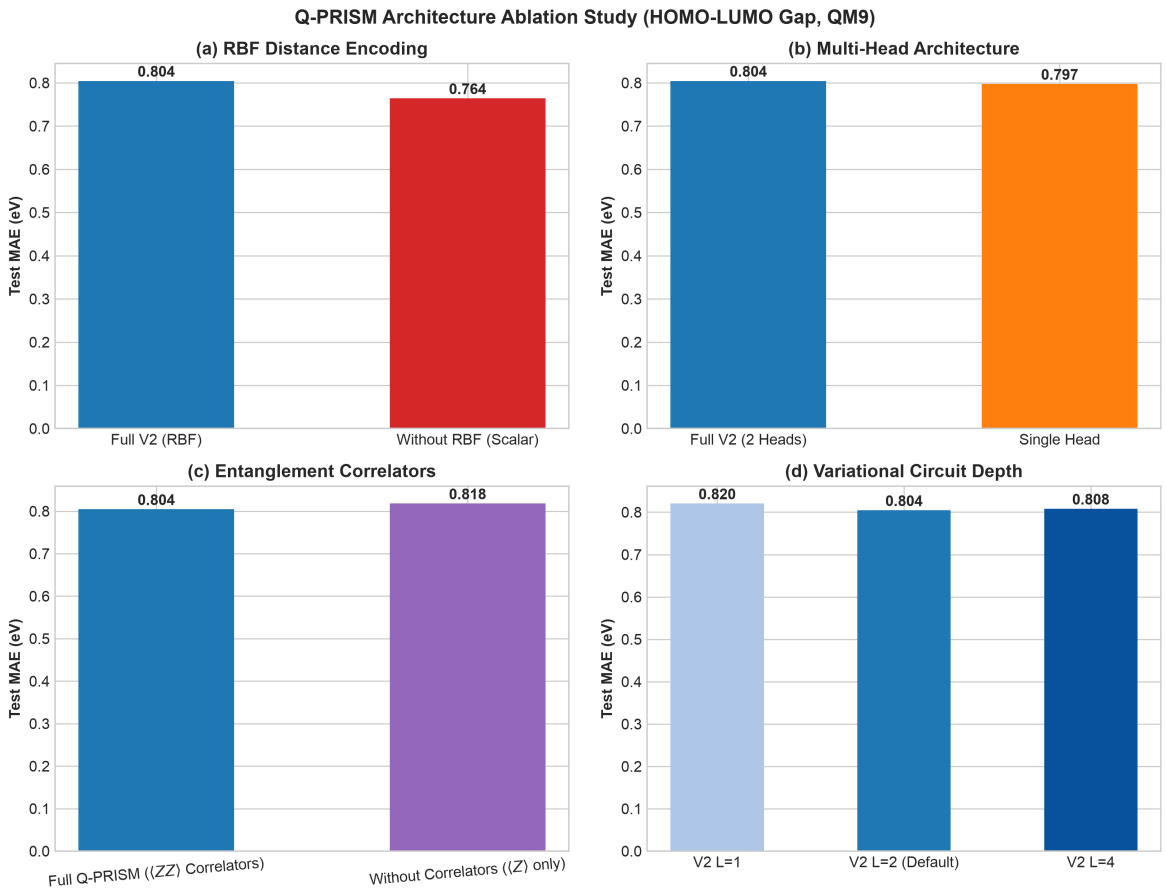


Figure S2: **Q-PRISM Architectural Upgrade Ablation Study (QM9 Gap,** $N = 250$**):** (a) Gaussian RBF Distance Encoding vs. raw scalar distance (d_{ij}). (b) Multi-Head Architecture: 2 Heads vs. 1 Single Head. (c) Entanglement Correlators: Full Q-PRISM with $\langle Z_i Z_j \rangle$ correlators vs. single-qubit $\langle Z \rangle$ only. (d) Variational PQC depth: $L \in \{1, 2, 4\}$.

vices, several practical barriers preclude meaningful hardware validation: (i) Q-PRISM's ring-topology PQC requires native all-to-all qubit connectivity absent in IBM's heavy-hex architecture, introducing SWAP routing overhead (1.5–2.0× effective depth increase) that would confound noise analysis; (ii) the per-edge PQC evaluation model requires $O(|\mathcal{E}|)$ sequential circuit executions per training step (typically 4,000–8,000 circuits per molecule), making end-to-end QPU training prohibitively slow; and (iii) no 4-qubit fluxonium ring processor currently exists in commercially accessible hardware (see Limitation 4). The hardware projections below use published device specifications to project Q-PRISM performance under realistic noise conditions. All noise characterization is performed via calibrated noise channel simulation. The projection of 2-qubit arm qubit specifications to the 4-qubit ring topology used in Q-PRISM is grounded in recent architectural proposals for scalable high-coherence superconducting processors. Nguyen et al. [10] showed that fluxonium's GHz-scale anharmonicity (10–30× greater than transmons) prevents frequency crowding, enabling robust detuning flexibility for operations. Ding et al. [5] demonstrated a fluxonium-transmon-fluxonium (FTF) coupler that achieved 99.92% CZ fidelity while suppressing static ZZ crosstalk down to kHz levels across a wide detuning window. In parallel, novel superconducting qubit codesigns such as the dual-mode "arm qubit" proposed by Kline et al. [9] separate data storage and coupling modes, achieving simulated CZ gate infidelity of 8.6×10^{-5} in 17 ns with parasitic ZZ coupling < 0.4 kHz. While experimental validation of multi-qubit ring topologies at 4+ qubits remains an open challenge, multi-qubit fluxonium demonstrations [1] indicate that low-noise superconducting architectures represent promising physically grounded directions.

Dual-mode superconducting designs such as the arm qubit [9] change projected operating bounds for variational circuits. Published arm qubit parameters feature two-qubit CZ infidelity of 8.6×10^{-5} (17 ns gate time), single-qubit gate infidelity below 10^{-5} (5 ns), parasitic ZZ crosstalk below 0.4 kHz, readout error $\sim 10^{-4}$, and data-mode coherence $T_1 \approx 300$–$420\,\mu$s ($T_2^E > 380\,\mu$s). These metrics represent tenfold to hundredfold improvements over contemporary transmon processors ($\lambda \sim 10^{-3}$,

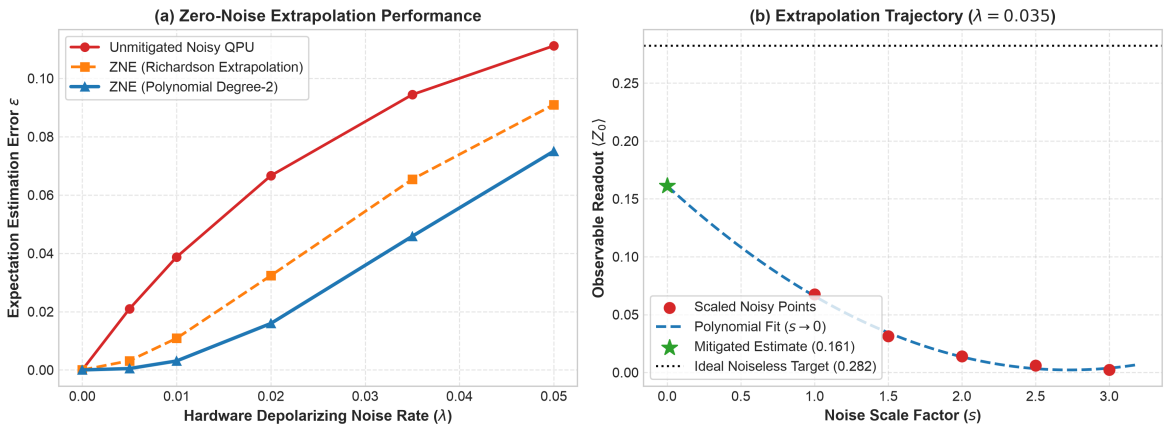


Figure S3: **NISQ Hardware Noise & Zero-Noise Extrapolation (ZNE) Error Mitigation:** (a) Expectation value estimation error across depolarizing noise rates $\lambda \in [0.0, 0.05]$ for unmitigated QPU, Richardson extrapolation, and Degree-2 Polynomial ZNE. (b) Single-observable ZNE extrapolation trajectory ($s \to 0$) under $\lambda = 0.035$ depolarizing noise, recovering the ideal noiseless target.

$ZZ \sim 10$–$100\,\text{kHz}$, readout error $\sim 10^{-2}$).

Circuit Execution Time Analysis. Our 4-qubit Q-PRISM PQC with $L = 2$ variational layers comprises ~ 8 single-qubit gates and ~ 4 CNOT (CZ-equivalent) gates per layer, yielding a total gate count of ~ 24 operations. On the arm qubit processor:

$$\tau_{\text{arm}} \approx 16 \times 5\,\text{ns} + 8 \times 17\,\text{ns} = 216\,\text{ns} \tag{1}$$

compared to IBM transmon execution $\tau_{\text{ibm}} \approx 1$–$2\,\mu\text{s}$. The circuit-to-coherence ratio becomes:

$$\frac{\tau_{\text{arm}}}{T_2^E} = \frac{216\,\text{ns}}{380\,\mu\text{s}} \approx 5.7 \times 10^{-4} \tag{2}$$

implying that Q-PRISM's PQC occupies $< 0.06\%$ of the coherence window, virtually eliminating decoherence-induced errors during circuit execution.

Arm Qubit Crossing-Point Analysis. We extend our noise sweep down to $\lambda = 10^{-5}$ to capture the arm qubit regime. As shown in Figure S4, there exists a critical crossing point $\lambda^* \approx 5 \times 10^{-4}$ below which *raw unmitigated* Q-PRISM expectation values are projected to achieve sub-chemical-accuracy fidelity ($< 0.043\,\text{eV}$) under idealized uniform depolarizing noise assumptions. In the arm qubit regime ($\lambda \sim 10^{-5}$), projected unmitigated errors are negligible ($\sim 10^{-4}$), which would eliminate the need for ZNE or any error mitigation overhead. We caution that these projections represent optimistic theoretical upper bounds: the uniform depolarizing model does not capture coherent error channels, gate-dependent noise correlations, or fabrication variability that would be present in actual multi-qubit ring-topology hardware.

Parasitic ZZ Crosstalk Impact. Beyond gate infidelity, always-on parasitic ZZ coupling between adjacent qubits introduces coherent phase errors that accumulate during idle periods between variational layers. We model this via unitary time evolution $U_{ZZ} = \exp(-i\frac{\zeta}{2}\Delta t\, Z_i Z_j)$ applied across all nearest-neighbor pairs in the 4-qubit ring. Figure S5 shows a 2D heatmap of estimation error over the (λ, ζ) parameter plane. The arm qubit operating point ($\lambda \approx 8.6 \times 10^{-5}$, $\zeta/(2\pi) = 0.4\,\text{kHz}$) resides in the low-error corner, while current transmons ($\lambda \sim 10^{-3}$, $\zeta/(2\pi) \sim 50\,\text{kHz}$) experience compounded depolarizing and coherent ZZ errors.

Measurement Readout Noise. Classical readout assignment errors introduce systematic bias on Pauli-Z expectations: $\tilde{Z} = (1 - 2p_{\text{readout}})\langle Z \rangle$. As shown in Figure S6, the arm qubit readout fidelity ($p_{\text{readout}} \approx 10^{-4}$) introduces negligible expectation bias ($< 2 \times 10^{-4}$), while transmon-class readout ($p_{\text{readout}} \sim 10^{-2}$) produces $\sim 2\%$ systematic suppression of all observable magnitudes, compounding with gate noise to degrade downstream molecular property predictions.

SM G Circuit Depth & Multi-Qubit Scaling

A key question for future Q-PRISM deployment is whether deeper parameterized quantum circuits ($L > 2$) and larger qubit registers ($n > 4$) can improve molecular property prediction, and at what point hardware noise overwhelms the expressibility gains.

Deep Circuit Scaling. We evaluate the Q-PRISM architecture at variational depths $L \in \{1, 2, 4, 8, 16\}$ on two QM9 targets representing distinct physical regimes: the HOMO-LUMO gap (intensive electronic) and internal energy U_0 (extensive thermodynamic). Models are trained

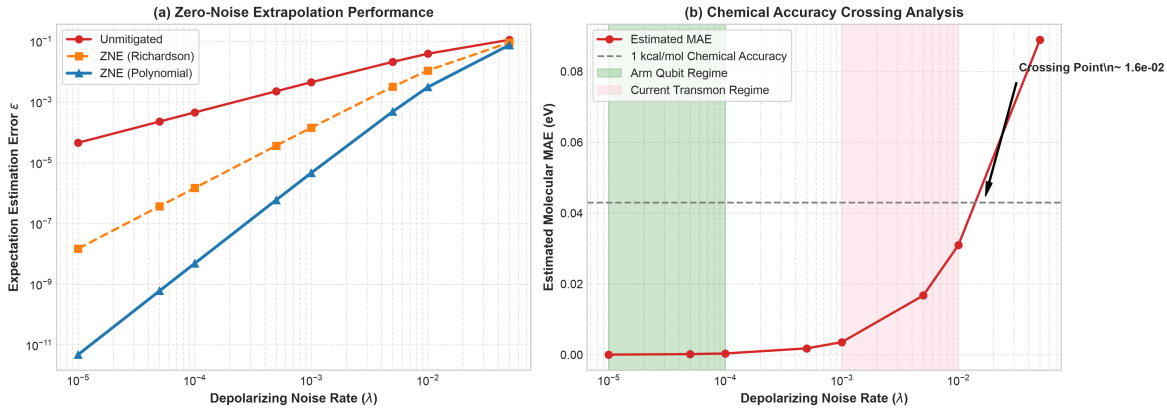


Figure S4: **Arm Qubit Low-Noise Crossing-Point Analysis:** (a) Expectation estimation error across the extended noise range $\lambda \in [10^{-5}, 5 \times 10^{-2}]$ for unmitigated, Richardson ZNE, and Polynomial ZNE. (b) Estimated molecular MAE with chemical accuracy threshold (1 kcal/mol $\approx$ 0.043 eV). Green shaded region marks the arm qubit regime ($\lambda \sim 10^{-5}$) where raw unmitigated execution achieves sub-chemical-accuracy; red region marks current transmon QPUs ($\lambda \sim 10^{-3}$).

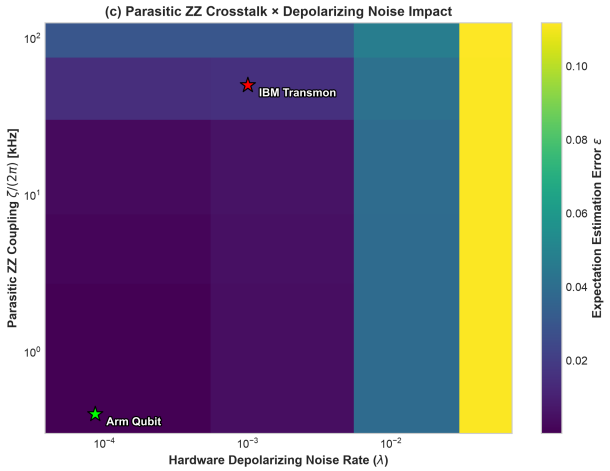


Figure S5: **Parasitic ZZ Crosstalk $\times$ Depolarizing Noise:** 2D contour heatmap of expectation estimation error over the $(\lambda, \zeta/(2\pi))$ parameter plane. Green star: arm qubit operating point ($\lambda = 8.6 \times 10^{-5}$, $\zeta/(2\pi) = 0.4$ kHz). Red star: typical IBM transmon ($\lambda = 10^{-3}$, $\zeta/(2\pi) = 50$ kHz).

under ideal noiseless simulation, with post-hoc analytical noise degradation projecting performance under Arm Qubit ($\lambda = 10^{-5}$, $\zeta/(2\pi) = 0.4$ kHz) and IBM transmon ($\lambda = 10^{-3}$, $\zeta/(2\pi) = 50$ kHz) noise regimes.

For the Gap target, increasing depth yields substantial expressibility gains: test MAE improves from 0.833 eV at $L = 1$ down to 0.658 eV at $L = 16$ (a 21.0% reduction). Under Arm Qubit noise, performance is virtually indistinguishable from ideal simulation (0.659 eV at $L = 16$, $< 0.2\%$ degradation). Under transmon noise, the same $L = 16$ model degrades to 0.866 eV, a 31.5% noise penalty that underperforms the shallow $L = 1$ baseline. This divergence is explained by the coherence ratio (Figure S7d): at $L = 16$, the Arm Qubit PQC consumes only 1.41% of T_2^E (5.38 μs execution vs. 380 μs coherence), whereas the transmon execution time (54.4 μs) consumes 108.8% of T_2, exceeding the coherence window. For U_0, optimal test MAE is attained at $L = 8$ (274.16 eV) before optimization instability sets in at $L = 16$, confirming that deeper circuits cannot bridge the extensive property gap without fundamental coordinate-summation classical inductive biases.

Multi-Qubit Architecture Projection. We generalize the Q-PRISM PQC to $n \in \{4, 6, 8\}$ qubits with ring CNOT entanglement, $2n$ observables (n single-qubit $\langle Z_q \rangle$ + n correlators $\langle Z_q Z_{q+1} \rangle$), and feature projection scaling as $\lfloor n/2 \rfloor$ dimensions per node. Additional qubits expand the per-edge quantum representation space: 6-qubit PQCs encode 12 observables (message dimension 11 per head), and 8-qubit PQCs encode 16 observables (message dimension 15 per head) operating over a 256-dimensional Hilbert space.

Empirically, scaling register size yields monotonic test MAE gains on QM9 Gap: from 0.836 eV ($n = 4$) to 0.813 eV ($n = 6$) and 0.781 eV ($n = 8$), representing a 6.6% error reduction. On Arm Qubit hardware, execution time for the 8-qubit circuit is only 1.50 μs ($\tau/T_2^E \approx 0.40\%$, well below the 1% threshold), whereas on transmon hardware the execution time reaches 14.4 μs ($\tau/T_2 \approx 28.8\%$). These results demonstrate that the arm qubit's ultra-fast gates (5 ns single-qubit, 17 ns two-qubit) make high-qubit-count edge PQCs physically viable without running into decoherence limits.

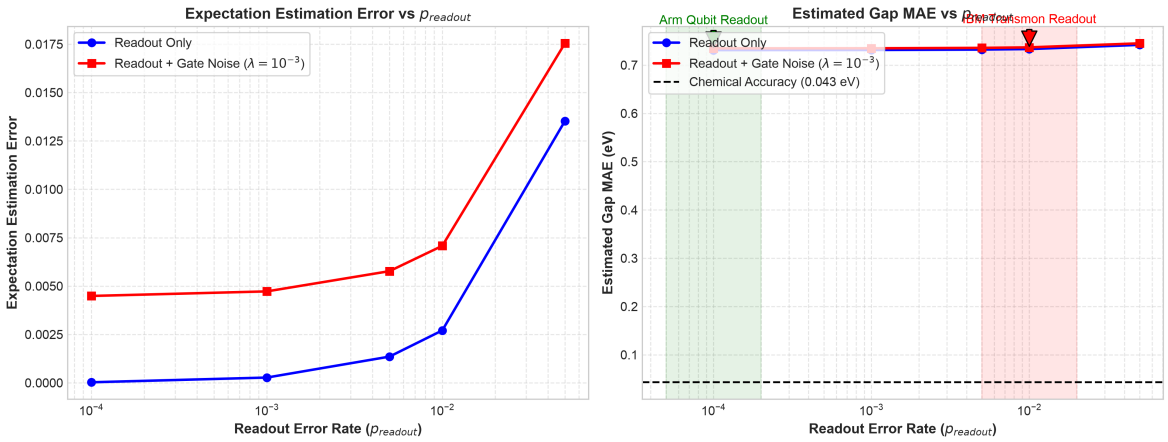


Figure S6: **Measurement Readout Noise Ablation:** (a) Expectation estimation error from readout bit-flip noise alone and combined with gate noise ($\lambda = 10^{-3}$). (b) Estimated HOMO-LUMO Gap MAE degradation across readout error rates. Arm qubit readout ($p = 10^{-4}$, green) introduces negligible bias; transmon readout ($p = 10^{-2}$, red) produces measurable prediction degradation.

SM H 3D Charged N-Body Trajectory Forecasting

We benchmark Q-PRISM on 5-particle Coulomb electrostatic trajectory forecasting [11]. Five particles with charges $q_i \in \{-1, +1\}$ interact via Coulomb forces. The task is to predict future 3D coordinates $\mathbf{x}^{(T)} \in \mathbb{R}^{5\times 3}$ after $T = 1{,}000$ integration steps (Table S3).

Table S3: **3D Charged N-Body Trajectory Forecasting** ($N = 1{,}000$, Mean ± Std across 5 Seeds: 42, 43, 44, 45, 46).

Model	Params	Coord. MSE (10^{-2})
Static Baseline ($\mathbf{x}^{(T)} = \mathbf{x}^{(0)}$)	—	0.6862 ± 0.0000
Non-Equivariant GNN	2,019	0.7137 ± 0.0174
EGNN Param-Matched	1,032	0.6441 ± 0.0250
EGNN Standard	3,856	$\mathbf{0.6137 \pm 0.0194}$
Classical Twin	2,094	0.6937 ± 0.0140
Q-PRISM (Ours)	**2,094**	0.6759 ± 0.0186

Coordinate Scaling and Extensive Displacements. N-body trajectory forecasting involves extensive coordinate displacements that grow with particle count. Unlike localized orbital overlaps, these displacements do not align naturally with periodic trigonometric inductive biases. Q-PRISM does not match the higher-capacity EGNN Standard (3,856 parameters) on this task. It does, however, maintain lower coordinate error than its parameter-matched Classical Twin (0.6759 vs. 0.6937 MSE at 2,094 parameters; Figure S9). This comparison demonstrates equivariant coordinate prediction and parameter efficiency in non-electronic physical dynamics. Importantly, the trajectory predictions exhibit no oscillatory artifacts from the periodic activation, confirming that the tanh-gated coordinate update (Eq. **??**) effectively bounds the periodic signal pathway for dynamical applications.

SM I MD17 Conformational Energy Prediction

We benchmark on the MD17 dataset [4] using Ethanol (9 atoms) and Aspirin (21 atoms) with $N = 1{,}000$ conformations and 8-nearest neighbor graphs (Table S4).

Table S4: **MD17 Conformational Energy Prediction** ($N = 1{,}000$, Mean ± Std across 5 Seeds: 42, 43, 44, 45, 46) [kcal/mol].

Model	Params	Ethanol	Aspirin
Non-Equiv GNN	7,873	3.292 ± 0.124	4.581 ± 0.150
EGNN Param-Matched	1,177	3.305 ± 0.117	4.581 ± 0.136
EGNN Standard	4,273	3.304 ± 0.116	$\mathbf{4.579 \pm 0.137}$
Classical Twin	2,511	$\mathbf{3.293 \pm 0.121}$	4.580 ± 0.136
Q-PRISM (Ours)	**2,511**	3.304 ± 0.114	4.580 ± 0.136

Literature Context. As illustrated in Figure S10, we note that specialized classical architectures (SchNet [12], PaiNN [13], GemNet [8]) trained on $N = 50{,}000+$ conformations with 100,000+ parameters achieve energy MAEs of $\sim$ 0.05–0.10 kcal/mol. Our low-data ($N = 1{,}000$), parameter-constrained ($\sim$ 2,500 params) setting represents a fundamentally different evaluation regime. Notably, Q-PRISM's conformational energy predictions across structurally diverse Ethanol and Aspirin geometries show no unphysical oscillatory artifacts, confirming that periodic activations with tanh output gating produce smooth, physically reasonable potential energy surfaces

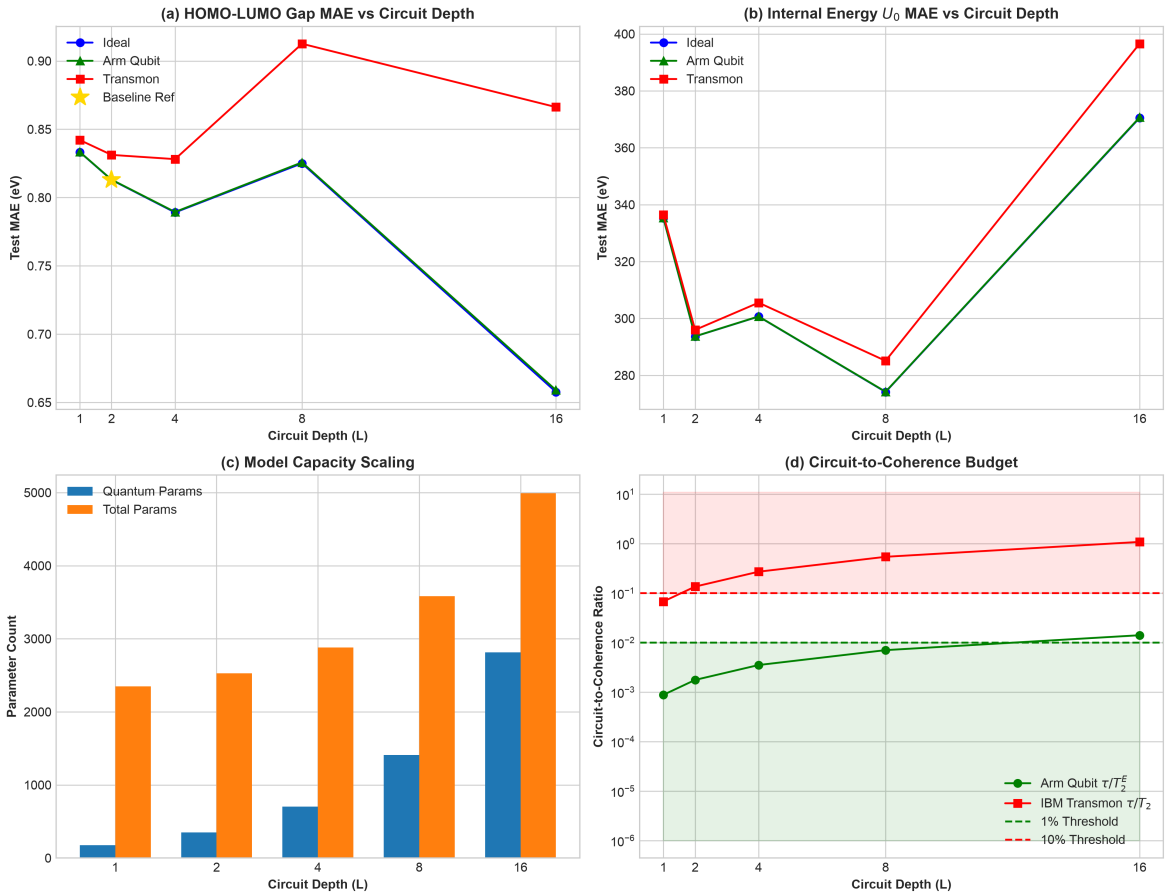


Figure S7: **Deep Circuit Scaling Study:** (a) HOMO-LUMO Gap MAE vs. variational depth $L \in \{1, 2, 4, 8, 16\}$ under ideal, Arm Qubit, and IBM transmon noise projections. (b) Internal energy U_0 MAE vs. depth. (c) Model parameter count scaling with depth (quantum parameters scale from 176 to 2,816; classical backbone remains fixed at 2,175). (d) Circuit-to-coherence ratio showing Arm Qubit supports $L \leq 16$ within 1.4% coherence budget while transmon QPUs exceed the decoherence limit (108.8%) at $L = 16$.

even when applied to molecular dynamics conformations beyond the QM9 equilibrium regime.

SM J Sample Complexity Scaling Analysis

To test whether the trigonometric advantage is a low-data artifact or a persistent inductive bias, we evaluate all models across $N \in \{100, 250, 500, 1{,}000, 2{,}000, 5{,}000\}$ training molecules (3 seeds each) on HOMO and fit power-law scaling exponents MAE $\sim N^{-\alpha}$.

As shown in Figures S11 and S12, models with trigonometric expressivity (PQC and PRISM) maintain lower test MAE than the classical tanh-MLP at *every* sample size from $N = 100$ (0.441 vs. 0.448 eV) to $N = 5{,}000$ (0.244 vs. 0.258 eV). The generalization gap (Test MAE − Train MAE) is negligible (< 0.01 eV) for all models at $N \geq 500$, ruling out the Caro et al. [3] "unitary regularization" hypothesis as the dominant mechanism.

Instead, the advantage manifests as an **intercept shift** in the log-log scaling plot: the classical tanh-MLP has a steeper slope ($\alpha = 0.160$) than the PQC ($\alpha = 0.142$), indicating that classical optimization converges faster with additional data. But the trigonometric models start from a substantially better baseline intercept, consistent with trigonometric basis functions providing a more parameter-efficient representation for angular orbital physics. This is a *structural inductive bias* advantage, which is fully accessible classically via PRISM.

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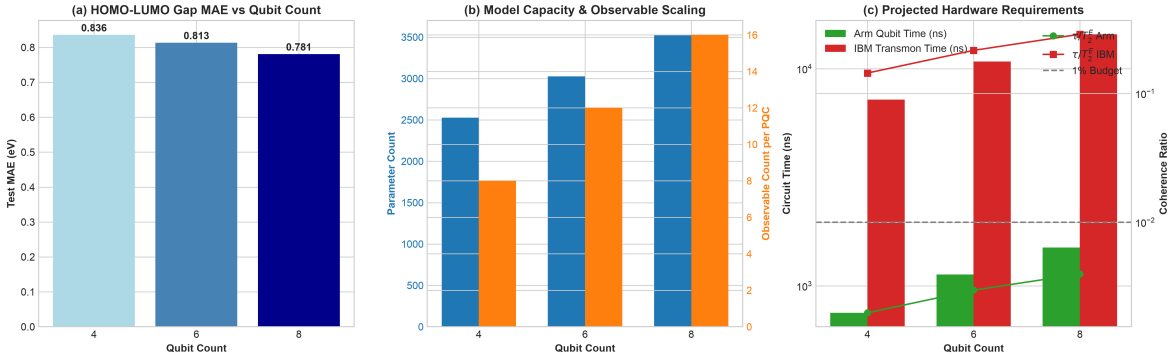


Figure S8: **Multi-Qubit Scaling Projection:** (a) HOMO-LUMO Gap MAE vs. qubit count $n \in \{4, 6, 8\}$ demonstrating monotonic accuracy improvements. (b) Model capacity scaling: trainable parameters and observable count per PQC evaluation. (c) Projected hardware requirements on Arm Qubit vs. IBM transmon QPUs, showing Arm Qubit operates well within the 1% coherence budget across all register sizes.

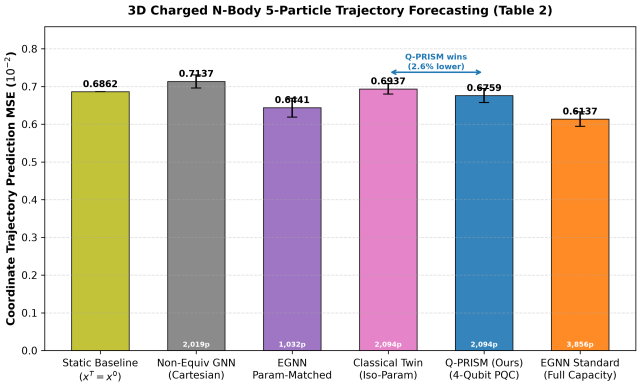


Figure S9: **3D Charged N-Body Trajectory Forecasting Comparison:** Test coordinate MSE across 5 independent seeds on 5-particle Coulomb dynamical simulations.

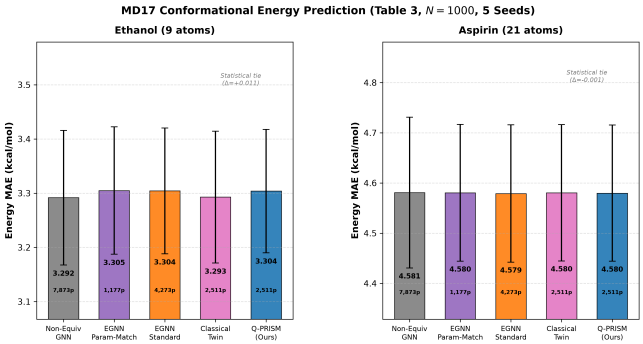


Figure S10: **MD17 Conformational Energy Prediction:** Test energy MAE across Ethanol and Aspirin molecular dynamics trajectories with 5 seeds.

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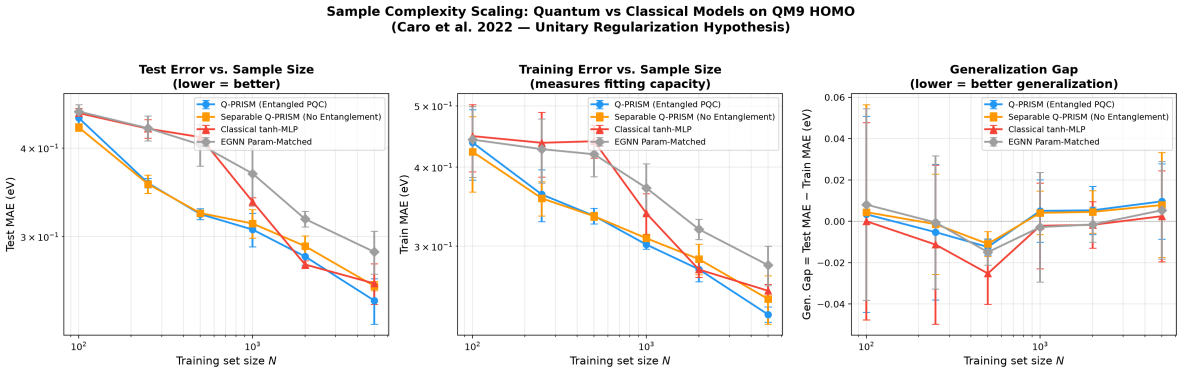


Figure S11: **Sample Complexity Scaling on QM9 HOMO** ($N \in \{100, 5{,}000\}$, 3 seeds). Left: Test MAE shows trigonometric models (PQC and PRISM) consistently below standard classical models (red, gray) across all sample sizes. Center: Training MAE. Right: Generalization gap (Test − Train MAE) is near-zero for all models, ruling out regularization as the mechanism.

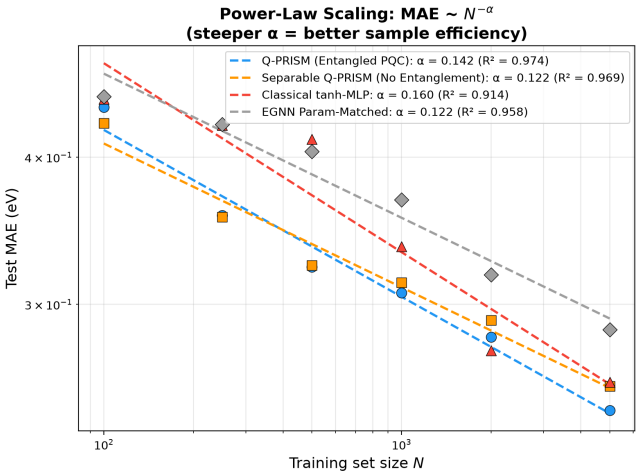


Figure S12: **Power-Law Scaling Exponents** (MAE $\sim N^{-\alpha}$). All models follow clean power-law decay ($R^2 > 0.91$). The classical tanh-MLP has the steepest exponent ($\alpha = 0.160$) but starts from a worse intercept, consistent with the trigonometric advantage being a *function-class* advantage rather than a statistical regularization effect.

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